PROJECT PORTFOLIO · REMOTE SENSING + MACHINE LEARNING

Iceberg Alley SAR

Finding icebergs off Newfoundland in Sentinel-1 radar with THOR, a SAR foundation model, benchmarked against a classical CA-CFAR detector.

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October 2026

Code: `github.com/natali-kovalski/iceberg-alley-sar`

Live viewer: `natali-kovalski.github.io/iceberg-alley-sar`



The live viewer, 8 May 2025: THOR detections (cyan rings), CA-CFAR baseline (orange dots), pack ice (blue), likely floes in grey.

84% of analyst-labelled icebergs found by THOR on four held-out frames (431 of 512)

74% found by the CA-CFAR baseline with the same number of detections

11 vs 4 large bergs (>120 m, of 14) found by THOR vs. the CFAR

28% fewer detections for THOR to reach the CFAR's 74% recall (≈670 vs 926)

Summary

I added the **THOR foundation model** (frozen encoder plus a small trained head) as a second iceberg detector next to my CA-CFAR baseline. On four held-out Sentinel-1 frames from April 2019 (512 analyst-labelled bergs), THOR found **84%** of the labelled bergs and the CFAR found **74%**, with both detectors allowed the same number of detections. THOR also found **11 of 14 large bergs** (>120 m), where the CFAR found 4.

Setup

The **baseline** is a cell-averaging CFAR on HV intensity (Sentinel-1 IW, HyP3 RTC at 10 m), with land and pack-ice masks and an HH co-pol check.

The setup follows Forgaard et al., *Efficient iceberg detection in Sentinel-1 imagery using the THOR foundation model* (IGARSS 2026). The article worked with VV/VH polarization; I used **HH/HV**, and the model accepted it well. I used the **tiny model** (FM4CS, Apache 2.0), which the article reported as performing well, with 8×8 patches.

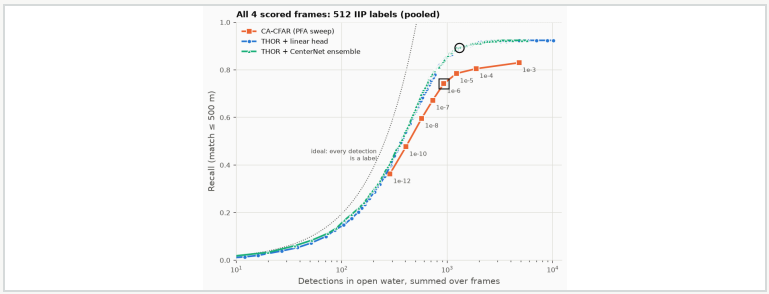
Data and labels

Training used **17 Sentinel-1 IW HH/HV frames** from May–June 2019 and April–May 2020. Each frame has same-pass IIP satellite-analyst labels, so no drift correction was needed. That gave **702 chips** of 512×512 px with **432 visible labelled bergs**. The chips were sampled around bergs, open water, the swath edge and the pack-ice edge.

- **Label cleaning.** IIP positions are truncated to 0.1 arc-minute. Moving them to the cell centre halved the position error (median 135 m to 67 m). I then snapped each label to the nearest bright HH peak within 200 m.
- **Ignored regions.** Labels on masked pixels and CFAR detections with no label are excluded from the loss. They are not treated as water.
- **Leakage.** Bergs drift only 10–20 km a day, so the same berg appears on nearby dates. All of 25 March to 6 May 2019 was held out as the test window, and head selection used year folds (train 2019, validate 2020, and the reverse).
- **Pack-ice set.** The Feb–Mar 2019 frames are mostly close pack ice. They were never trained on and are kept as a separate stress set.

TOOLS

`PyTorch · TerraTorch · Python · NumPy · SciPy · rasterio · GeoPandas · HyP3 · Vue 3 · CesiumJS`



Recall vs. detections, 4 held-out frames, 512 bergs: THOR linear (blue) and CenterNet (green) vs. the CA-CFAR over its false-alarm rates (orange).

Results on the held-out test frames

The test used four April 2019 frames with same-pass analyst labels. Nothing in the models or the comparison rules was tuned on them. Without AIS, precision can't be measured, so I compared **recall at a matched number of detections**. The CFAR's count at PFA 1e-6 was fixed, and THOR got the same number of top-scoring peaks. Both detectors used the same masks, labels and 500 m one-to-one matching.

- **THOR is ahead on every frame.** Pooled, it reaches the CFAR's recall with about 28% fewer detections. The CFAR saturates at 0.83 even at PFA 1e-3 (about 4,800 detections); THOR reaches that with about 900.
- **Large bergs (>120 m):** THOR 11 of 14, CFAR 4 of 14. The CFAR does detect them, but their grown blobs (340–470 m) exceed the length rule that rejects sea-ice strips, so they are thrown away.
- **Medium bergs (61–120 m)** carry most of the gain: 0.76 to 0.87 on 425 labels.
- **Small bergs (15–60 m)** are a tie at 0.71. They are near the ~20 m true resolution, and one THOR token is 80 m.
- **The matching radius doesn't explain the gap.** From 200 m to 500 m both curves are flat.

Limitations

The result is consistent, but the sample is narrow. The pooled gap (10 points) is about five times the binomial noise at n = 512. Still, the evidence is one month of one year, the frames share some bergs, and near range (<35° incidence) is untested.

- **Pack ice.** On its own, THOR fires on pack ice: 34–82% of its peaks on icy frames fall inside the CFAR's ice mask. On the pack-ice stress set its F1 is only 0.2–0.35. It depends on the CFAR pipeline's HV ice mask, and bergs within 5 km of pack ice are not scored for either detector.
- **Precision is unknown** for both detectors. At the matched count, the CFAR has 546 unlabelled detections and THOR has 495. Without AIS, ships and missed bergs can't be told apart.

Data: modified Copernicus Sentinel data (2019–2026) · International Ice Patrol (NSIDC G00807) · North American Ice Service · THOR: FM4CS (Apache 2.0), Forgaard et al., IGARSS 2026 · © OpenStreetMap contributors